

Top Slicing

27/01/2020

Applicability 1

- *Top Slicing is a caving method, causing subsidence at surface.
- *For top slicing, disturbance & caving of surface are to be permitted.
- *Under certain deep mining conditions, surface subsidence might not occur for a long time, but should be anticipated eventually.
- *Top Slicing is not a mass mining method.
- *

Applicability 2

*Top Slicing is an under hand method.

5A*Top slicing method can be used for extraction of pillars between stopes previously mined by some other method.

5A*Top slicing can also be used for robbing operation in caved or badly broken ground.

5A*When ground is so badly broken, then it is unsafe or impractical to mine upwards by any of overhead stopping method.

*In this condition, ore can be mined safely & at reasonable cost from top down wards by top slicing.

4A 4A *

Ore Body 1

1A*Ore body for top slicing should be fairly wide ore body, width of ore is 3m or more.

*Rock formation of Hang wall and ore is weak.

1A*Ore body is too weak to stand with out support over other than short span.

1A*Ore body has weak capping which will cave readily when support is removed & fill excavated space tight.

*Top slicing is generally deployed when ore body is steeply dipping.

Ore Body 2

- 1B*For secure full top slicing, top weight or vertical pressure is necessary.
- 2A*Side wall may be either weak or strong.
- 2A*If ore occurs in form of a narrow dipping body, a weak hanging wall is desirable for top slicing.
- *Top slicing is applicable to thin flat deposit or ore deposit of any shape.

Ore Body For Top Slicing 113

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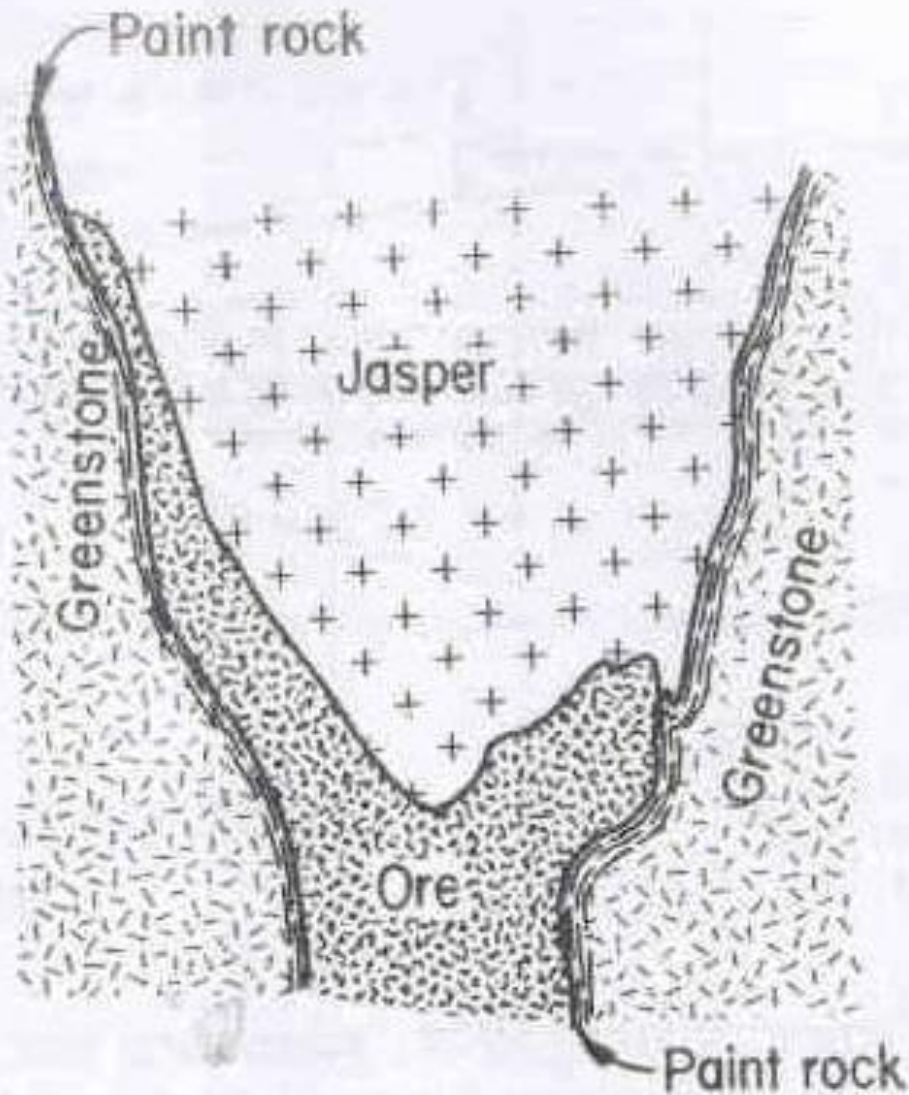


Fig. 16. Typical cross section at the Vermilion Range ore body.

Exploration 1

30A* Mine development plan are based on previous drilling exploration from surface, drilling of vertical holes.

30A* Normally vertical holes are drilled at 100m square.

30A* Closer drill hole spacing would permit better planning of development & improved economy.

30A* Drill hole drilling at 50m interval (in square pattern) would provide much better result.

30A* Some time additional holes are drilled between drill holes drilled earlier, for better specific information & purpose.

Development 1

31A *Shaft is usually sunk below low point. (bottom of ore body) (below deepest point of ore body).

31A*Shaft is normally located outside of ore body & centrally located between two ends of ore body.

D1*Main level / haulage level are used to haul ore to mine shaft for further hoisting to surface.

D1*Haulage level & shaft consist preliminary development of mine.

32A*Main haulage drift were driven from shaft & kept close to orebody, below bottom contours of ore at motor haulage grade.

Development 2

32A*Permanent development is designed to be outside of ore body, as much as in waste rock.

*In deep ore body, haulage way is driven through orebody and from it a two compartment timbered raise containing an ore chute and a man way is constructed throughout the entire height of the block up to overburden (capping).

32B*

9A*Total development cost is much higher than other contemporary method.

Development 3

32B*Main haulage drift were driven in such a way, so that under tramming drift are one above another.

*Haulage drift & tramming drifts are connected in such a frequent interval, way, so that tramming distance in sublevel gets reduced

9A*Considerable preparatory development is required prior to beginning of slicing operation.

Development 4

9A*Ore production from development heading is 20 to 25% higher than stoping.

16A*Period of development prior to production from slicing is relatively long.

16A*Rate of development has to be kept much in advance to obtained rated production from top slicing.

Development For Radial Slicing113 *

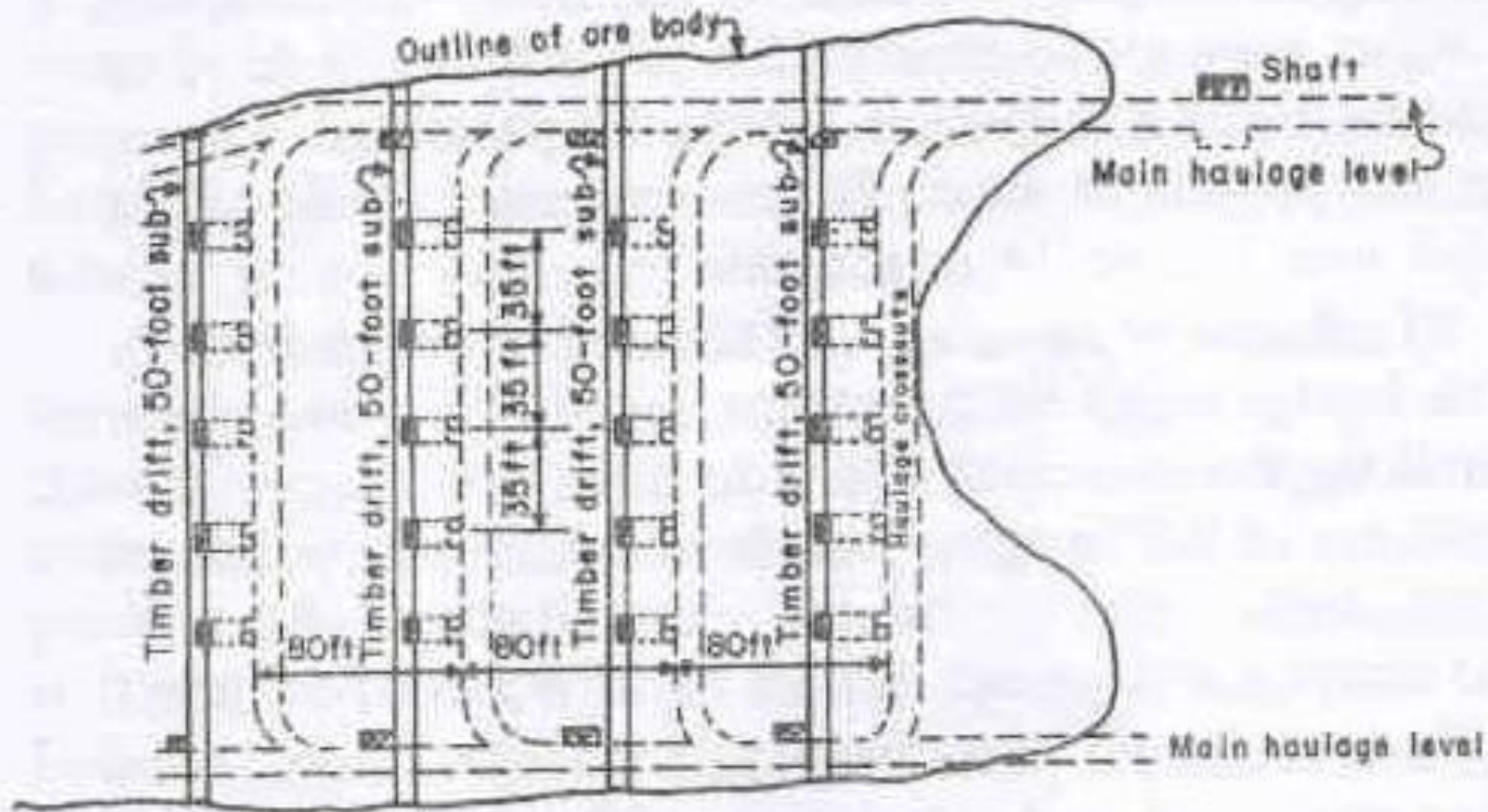


Fig. 17. Plan showing scheme of main and 15.2-m (50-ft) sublevel development in wide ore for radial slicing. Metric equivalent: $\text{ft} \times 0.3048 = \text{m}$.

Development 5

- *Vertical height of stopes is same as that of haulage level interval.
- *All stope ore pass leads to main drive for further hauling of ore to main ore pass for hoisting to through mine shaft.
- *Raise is taken to top sub level from haulage level.
- *Number of drift, cross cut at are excavated at top most sub level.
- *32A*Raises are put up from haulage drift at intervals of about 15m to top of ore.
- 32A*This raises are connected with timber drift on each sublevel, starting at top sublevel.

Development 6

32B*Parallel tramming drifts were driven approximately at right angles to timber drift on 15m center to match raises. Fig 8

32B*This drifts are extended up to margin, shore line or boundary of property.

33A*Some time, short transfer raises were put up & sub-drift continued from raises at a higher elevation.

33A*Ore from upper sublevel has to be transferred through raises to main haulage level.

*Number of drifts and cross-cuts are excavated at same distance below the top sub level.

Development 7

32B*Sublevel interval is 2 to 3m. (slice thickness).

33A*When tramming drift on upper level intercepted with bottom rock, with out shore line or boundary.

*Stope block divided in to horizontal slices of 2-3m thick.

72B79*Due to high timber consumption, separate shaft is made for men & timber handling.

72B*Normally haulage levels are excavated at 30 to 50m interval.

72B*Often, compartmental raise with large cross section are excavated for chute, men & timber supply.

Stoping 1

- *Ore is mined out as horizontal slices by drilling and blasting beginning at the top of the orebody.

- *Ore slice is removed in between capping (over Burdon) & level below.

- *Ore slice removed is normally 2-3m thick.

9A*Once slicing operation started, considerable tonnage of ore is produced.

A1*In top slicing, slice back is supported with help of timber with capping after drilling & blasting.

A1*After completing slice, back of slice is caved down by bringing down wooden support erected for this slice in floor.

Stoping 2

- *A2* Before bringing down capping, timbered mats is spread out on the floor of first slice (timber mat, i.e thin wooden logs and thick wooden planks tied together by wire) spread on floor of slice.
- *Immediately beneath the caved overburden, ore from first slice is being taken out, timber supports are erected on the floor of the slice.
- A2* Next lower slice is mined up under over laying mat & gob.
- A2* Accumulation of timber from upper slice & broken capping follows, as mining is moving progressively down wards. A2* Finger 1

Stoping 3

* B1*only one slice can be mined at one time.

B1*There has to be series of timber slice one beside another.

B1*As timber supports are blasted down, capping comes down along with already caved back (waste rock & timber remnant).

**The next slice of the orebody is then extracted by drilling and blasting working under the caved overburden resting on the timber mat. The slices of thickness 2-3m thick are extracted in descending manner

Stoping 4

*C1*Ore does not break by itself by caving action, induced by under cutting.

C1*Ore slice is broken down by conventional drilling & blasting.

C1*Only timber capping is broken down by caving action.

F1*Sublevel is above main haulage level, on which slicing operation is conducted.

F1*Sublevel may not be connected directly to shaft but they are connected with main haulage level by series of chute & man way entries.

Stoping 5

G1 In top slicing, stoping is termed as slicing.

G1* This slice is a timber supported drift driven under & up to capping or caved mat.

G1* Slice in length extends from raise, opening set or other starting point to merging of ore body, caved ground or predetermined mining limit.

G1* In sublevel caving, it is a timbered drift from which caving is conducted on retreat.

Stoping 6

- *H1*Tramming drift is developed for ore tramming in sublevel for dumping ore in to chute.
- H1*Timber drift are sublevel drift connecting a line of raises or crosscut & uses principally handling timber & supplies in sublevel.
- J1*Stope development is slice preparation for drilling & blasting.
- J1*It consists of man way from raises to sublevel, tramming & timber drift.
- I1*Opening set / open set is applied to first set of timber at entrance to slice & is key set of slice.

Stoping 7

*17A*Timbering & covering down slice takes long time of shift.

17A*Available time for breaking & mucking is much less.

*20A*This method can not adopt to sorting of waste in stope or to leave horses of rock in palce.

23A*Panel or block top slicing, in which, ore body is divided into a number of blocks or panels. Each having a central chute & manway raise.

23AFig – 1,2,6

Stoping 8

41A*It is usual for a gang of miners (two men) to drill, blast, scarp out & timber one set of round 1.5m per shift.

Classification Top Slicing

21A*Top slicing can be classified as flat or horizontal slicing & incline slicing.

21A*This depends on whether slice are driven level or incline from horizontal.

21A*Inclined method was evolved to eliminate hand shoveling by driving slice at such a angle, that broken ore would run to chute by gravity.

22A*Flat & horizontal slicing method can be in parallel direction or radial direction.

21A Fig 1,2,3

Parallel Horizontal Top Slice 1

22A*In parallel slicing, ore may also be scrapped directly to chute, but in other instance, it is scrapped in to cars.

22A*Scrapping in to car is generally used for slightly dipping, wide or thin ore bodies, where development in rock requires for scrapping direct to chute would be prohibited.

22A Fig – 1,2,4,5

23A*In top slicing, ore can be hand loaded to cars, scrapper loading to car, scrapping directly to chute & gravity handling to chute (incline top slicing).

Parallel Horizontal Top Slice 2

23A*Long wall slicing, in which a series of slicing are driven end to end from number of suitably arranged tramming drift, starting at margin of ore body, with caving & retreating from them.

22A*In parallel slicing, ore can be (1) scrapped directly to chute (2) scrapped in to cars (3) manually loaded to car, trammed & dumped in chute.

*

*

Parallel Horizontal Top Slice 3

34A*For mining thin, flat ore body, main haulage is driven in bed rock.

34A*Ore is scrapped in to car on sub level & hand trammed to chutes.

34AStoping by 3 slices

34A*In case of 3 slices, parallel slicing is followed for entire thickness of ore body.

34A*Scrapper loader loads car in sublevel.

34A*Car is hand trammed to chute in sub level.

Parallel Horizontal Top Slice 4

- 36A*In parallel top slice, slice usually starts on any given sublevel at property line or ore margin.
- 36A*Face is retreated in long wall fashion.
- 36A*This slices are driven at right angle to tramming drift or timber drift.
- 36A*This process is also known as Right angle slicing.
- 36A*Sublevel intervals in rectangular slicing varies from 3 to 4.5 (thickness of slice).
- 36A*When ore body is 4 to 6 m thick, it often taken as high slice from floor to capping in one operation

Parallel Horizontal Top Slice 5

- 36B*Each slicing is started by taking up a “Opening Set”
- 36B*Open set stands right angle to drift set.
- 36B*Drifts are usually driven smaller in size.
- 36B*Opening set goes up to capping or mat, as case may be.
- 36B*Back of drift support, slice has to be taken to make room for open set.
- 36B*From this opening set, slice is advanced in one direction to cave or to limit of slice.

Parallel Horizontal Top Slice 6

36C* In some cases, slice in only one direction, starting from drift slicing is carried out to cave or to next tramming drift. (Fig2)

36C* Other practice is slice is driven in both directions, by only half way, through pillar between tramming drift to muck slice from adjacent drift.

48A* In parallel slicing, ore is commonly loaded in to car & then trammed to chute.

48A* Right angle turn & sharp bend can be negotiated by scrapper, generally with use of deflecting sheave for rope, roller or boards nailed to bottom of post on corner.

Parallel Horizontal Top Slice 7

48B*This shall ease scrapper around them & prevent it catching on timber.

48B49*By close spacing of raise on sublevels, parallel slicing can be used for direct scrapping to chutes.

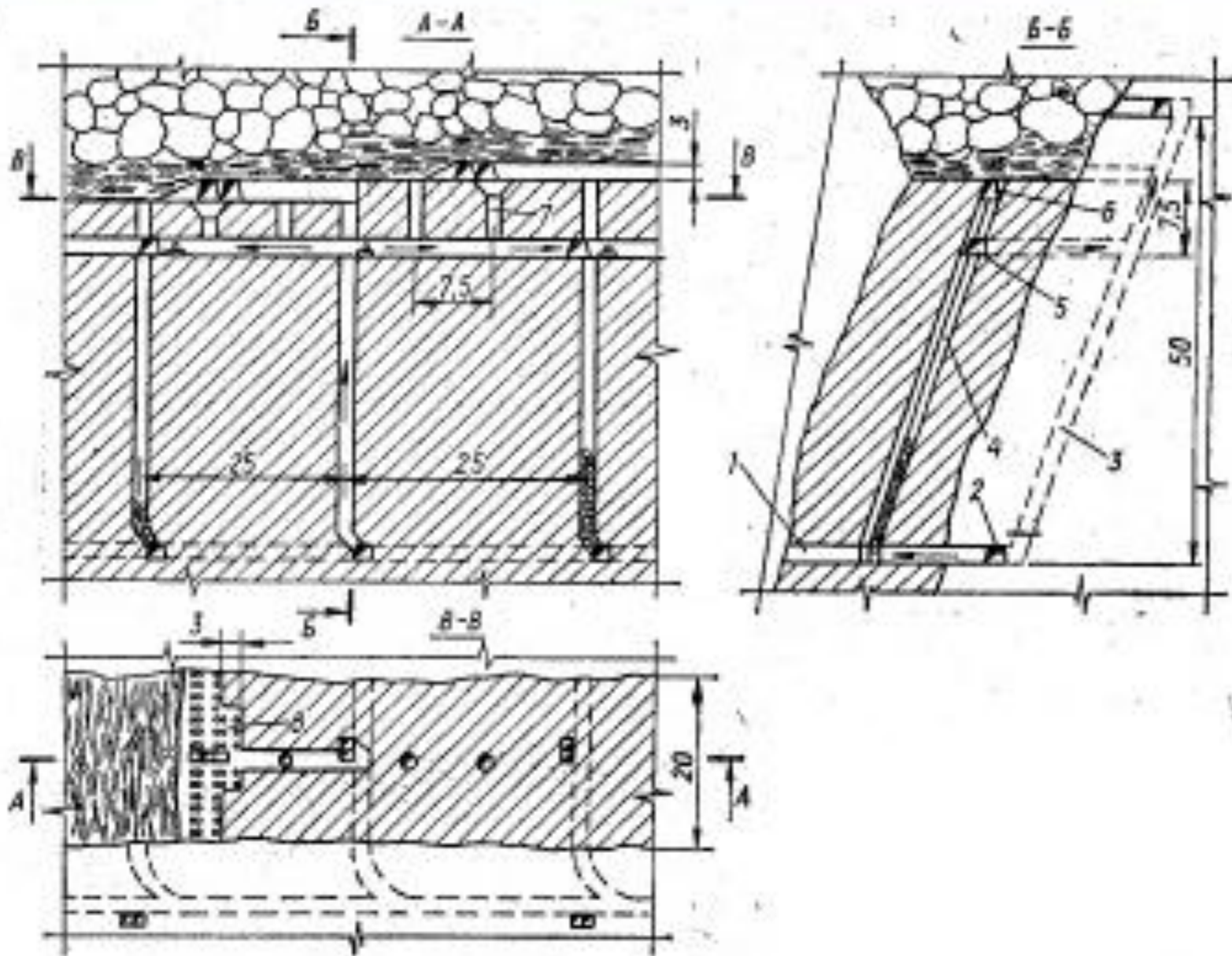
48B*Raises are usually about 5 to 11 m apart.

48B*Some time closer spacing is achieved by excavating raise branch from main raise (finger raise). Fog – 9.

48B*After first slice happened, second slice is started by propping cap on first set beyond opening set.

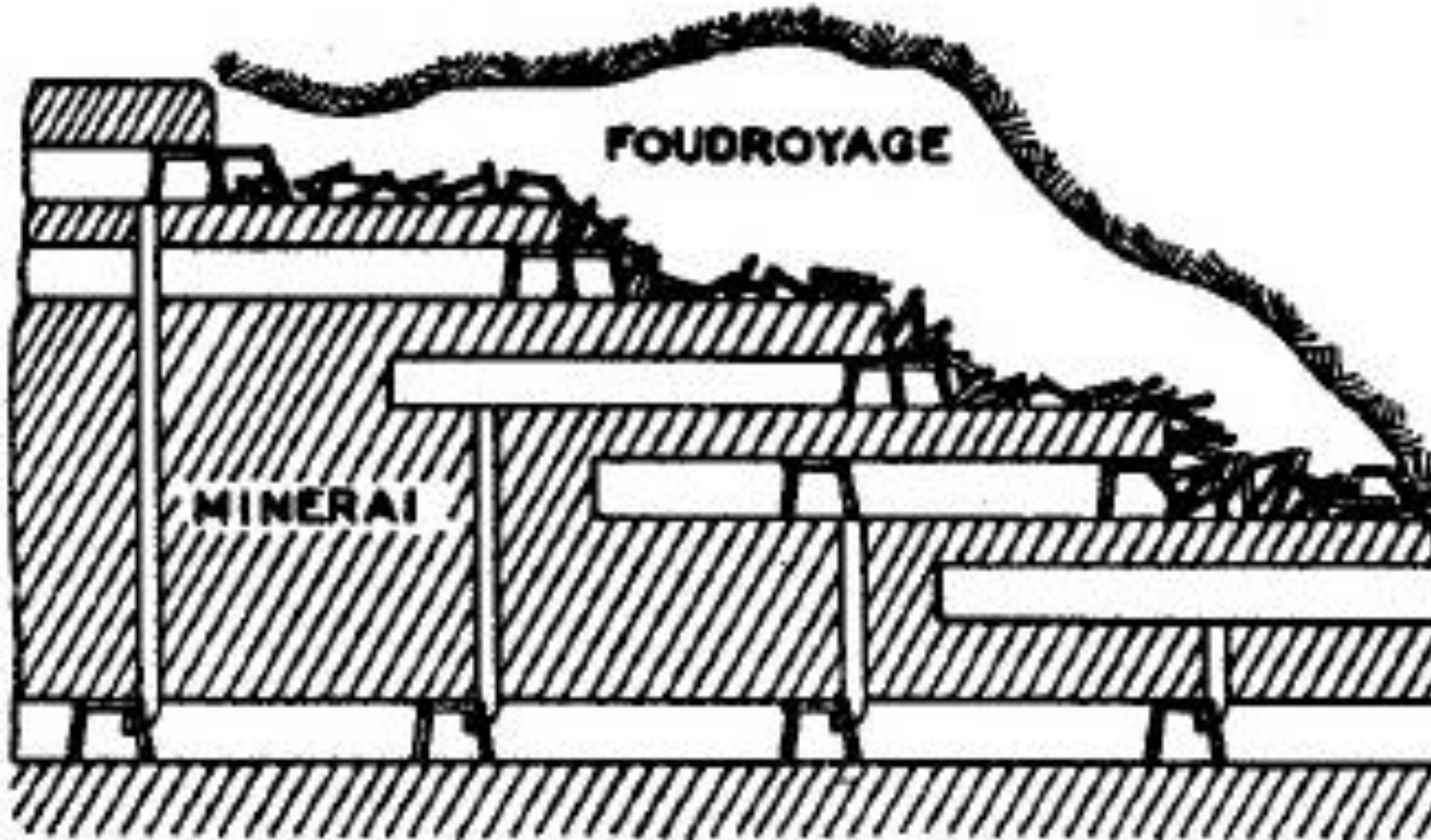
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Horizontal Parallel Top Slicing I

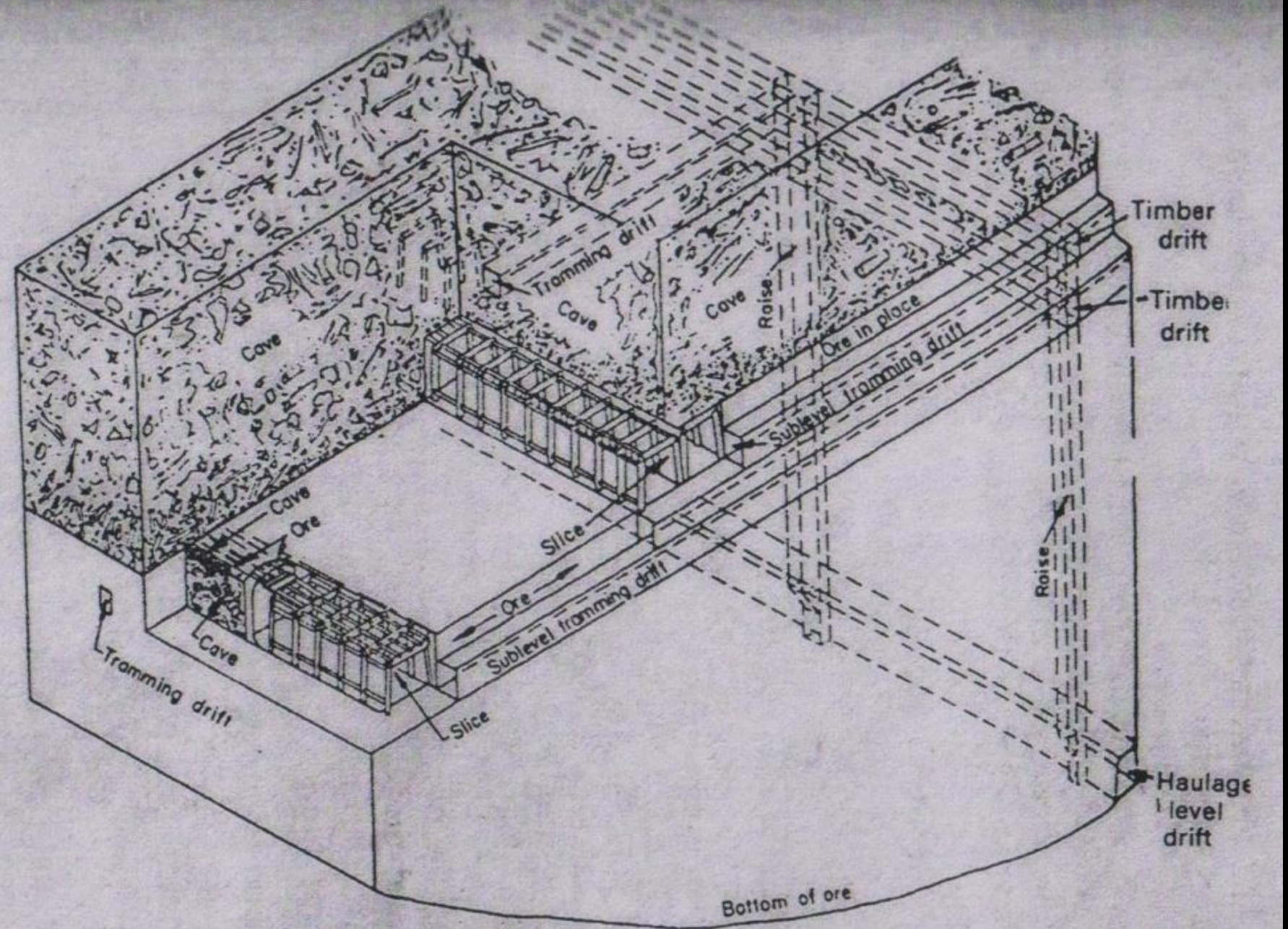


Horizontal Parallel Top Slicing

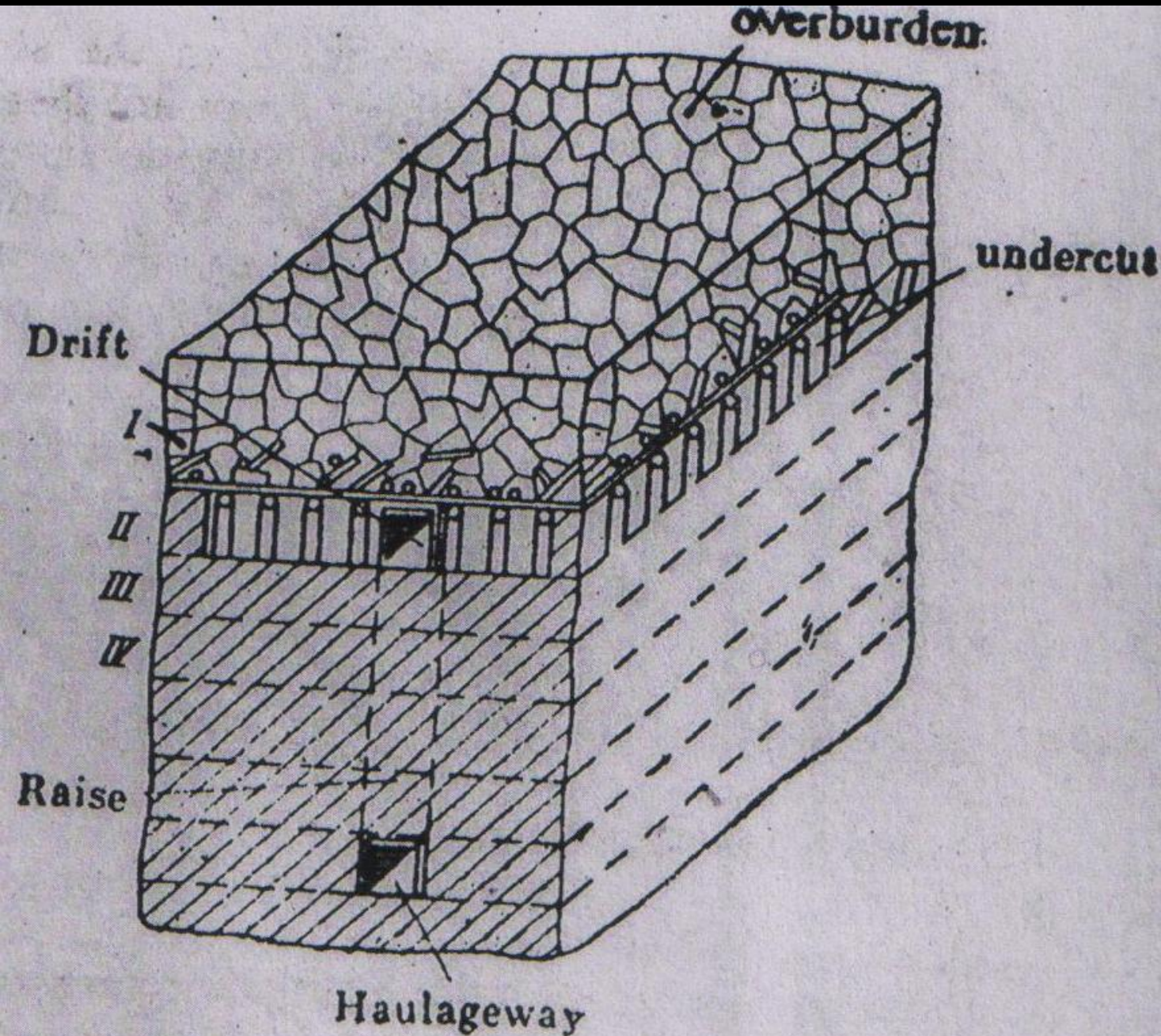
*Slices in staggered pattern.



Horizontal Parallel Top Slicing*



Horizontal Parallel Top Slicing



*Multiple slice thick ore body.

Top Slicing Thin Ore Body

- *Horizontal Parallel Top Slicing.
- *One slice thick ore Body.

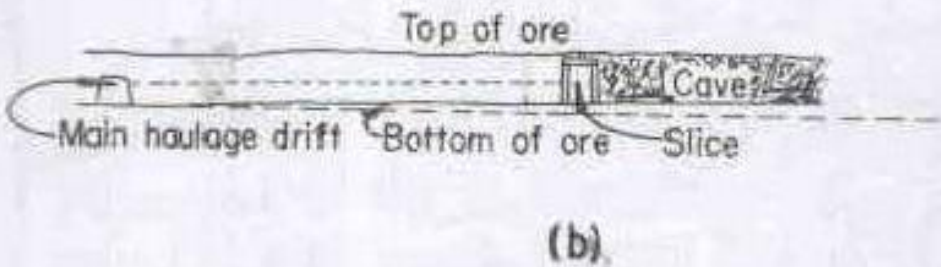
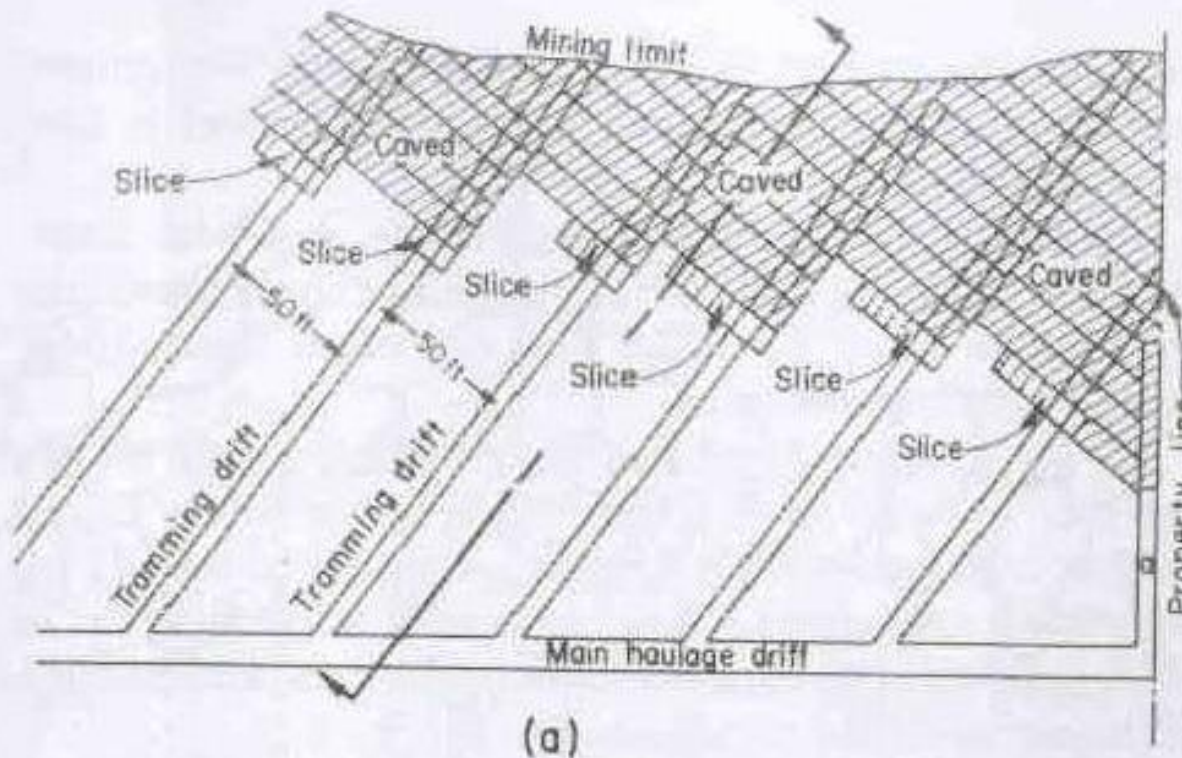
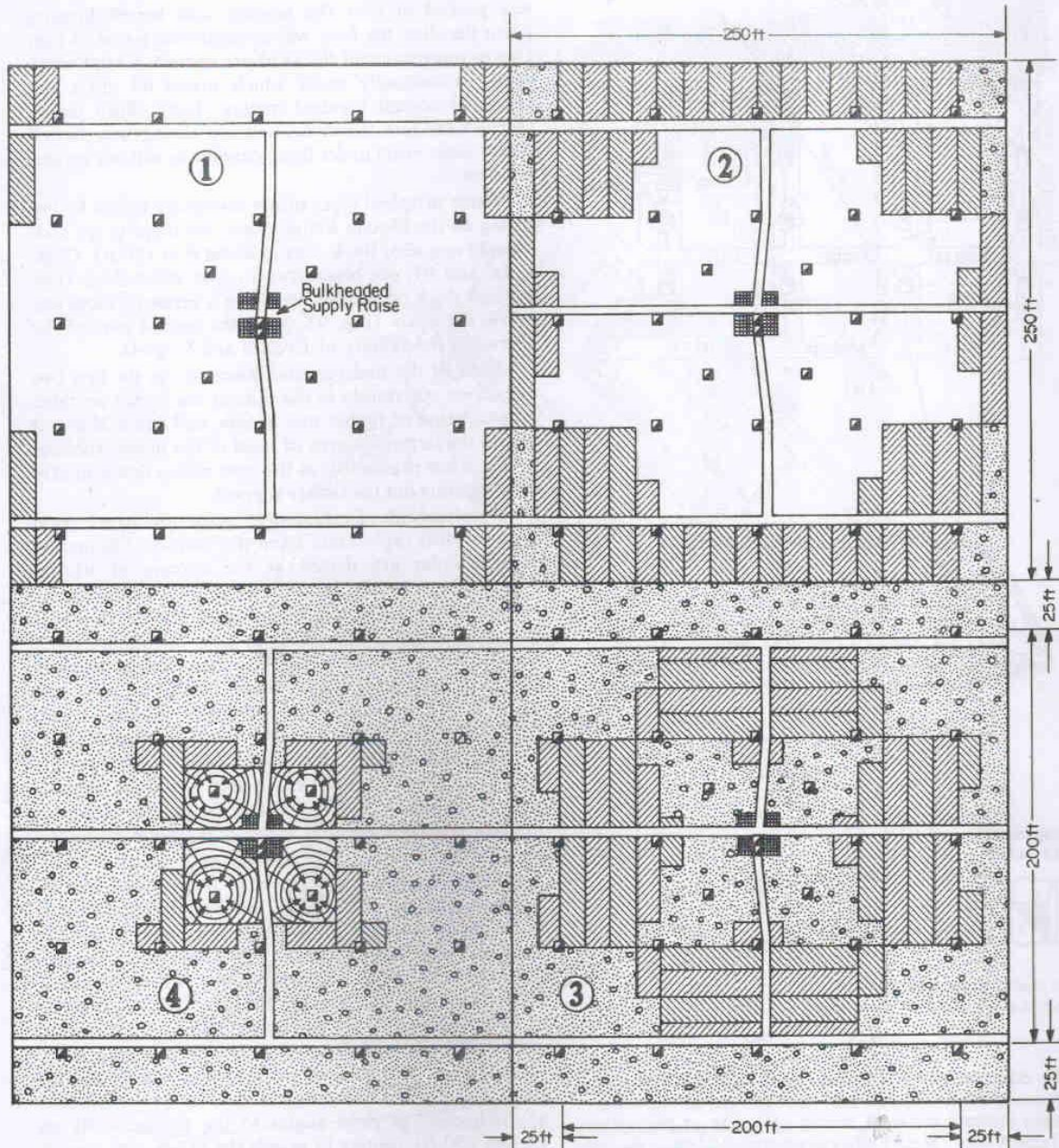


Fig. 2. Top slicing in a flat ore body one slice thick:
(a) plan; (b) section X-X. Metric equivalent: $\text{ft} \times 0.3048$
 $= \text{m}.$

Parallel Horizontal Top Slice 13



1. Haulage drift, chute & supply raise are developed.
2. Parallel Horizontal Slicing (first slice) starts, ore dumped in to raise.
3. Completion of 1st level slice.
4. Starting of next level slice.

Fig. 6. Plan of successive steps in block method of top slicing. Metric equivalent: $\text{ft} \times 0.3048 = \text{m}$.

Parallel Horizontal Top Slice 14

Parallel Horizontal Top Slicing in thin ore body with haulage level in waste rock.

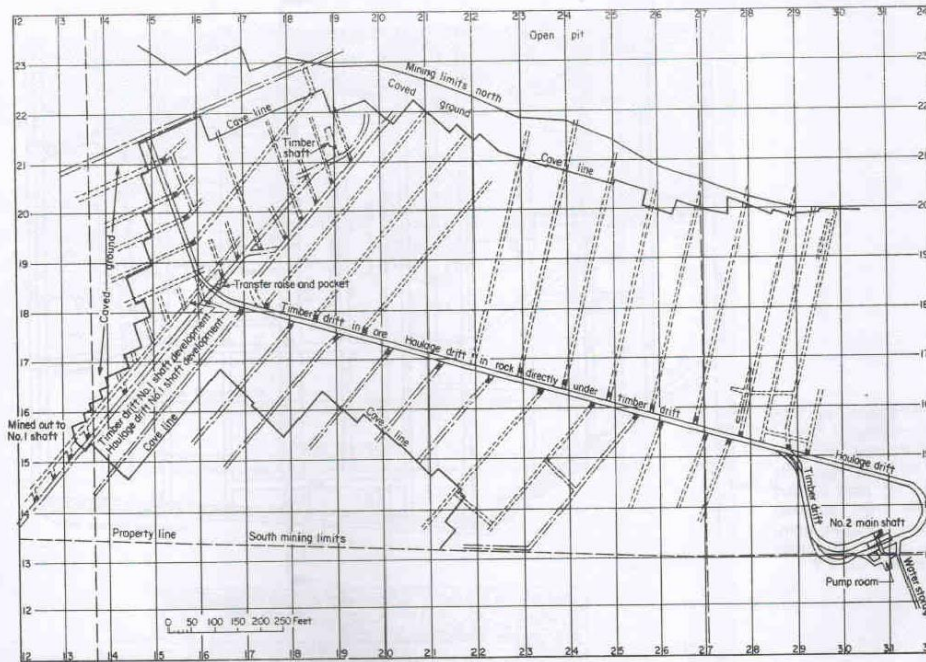


Fig. 7a. Plan of development for a thin ore body. Metric equivalent: $\text{ft} \times 0.3048 = \text{m}$.

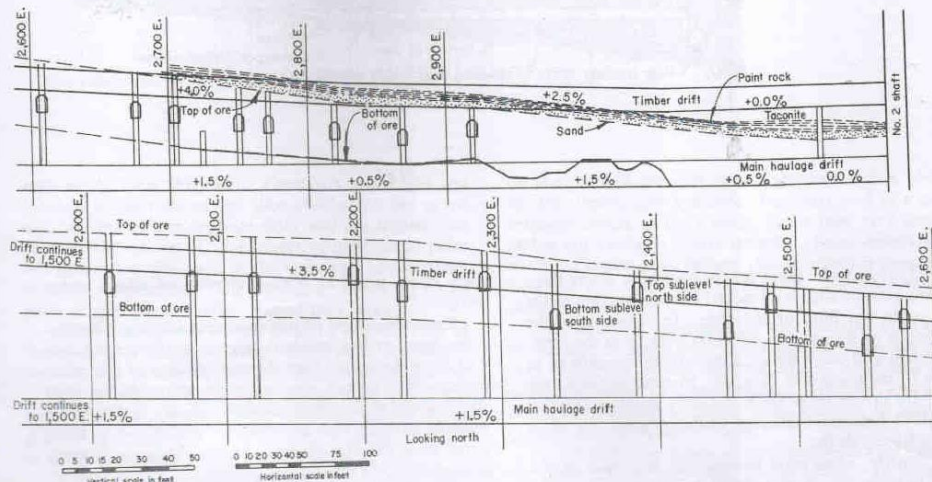


Fig. 7b. Section of thin ore body showing main haulage level in rock below ore. Metric equivalent: $\text{ft} \times 0.3048 = \text{m}$.

Parallel Horizontal Top Slicing17

Ore body is up to
4 slice thick.
Ore is scrapped
in to car.

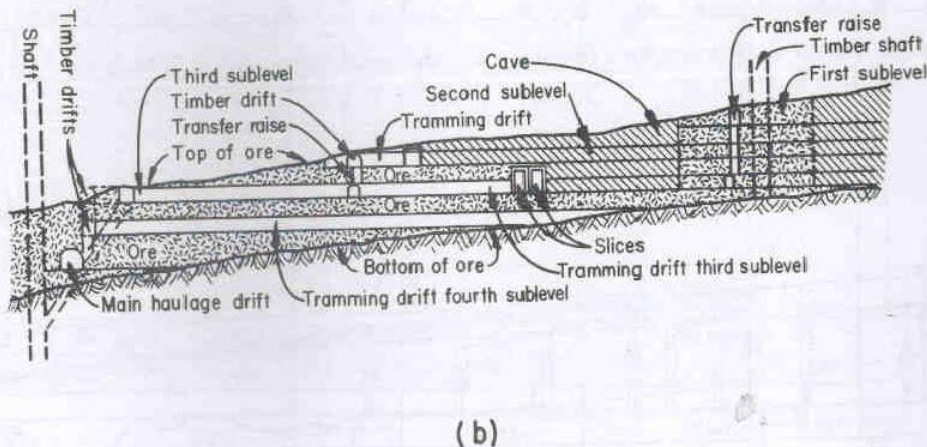
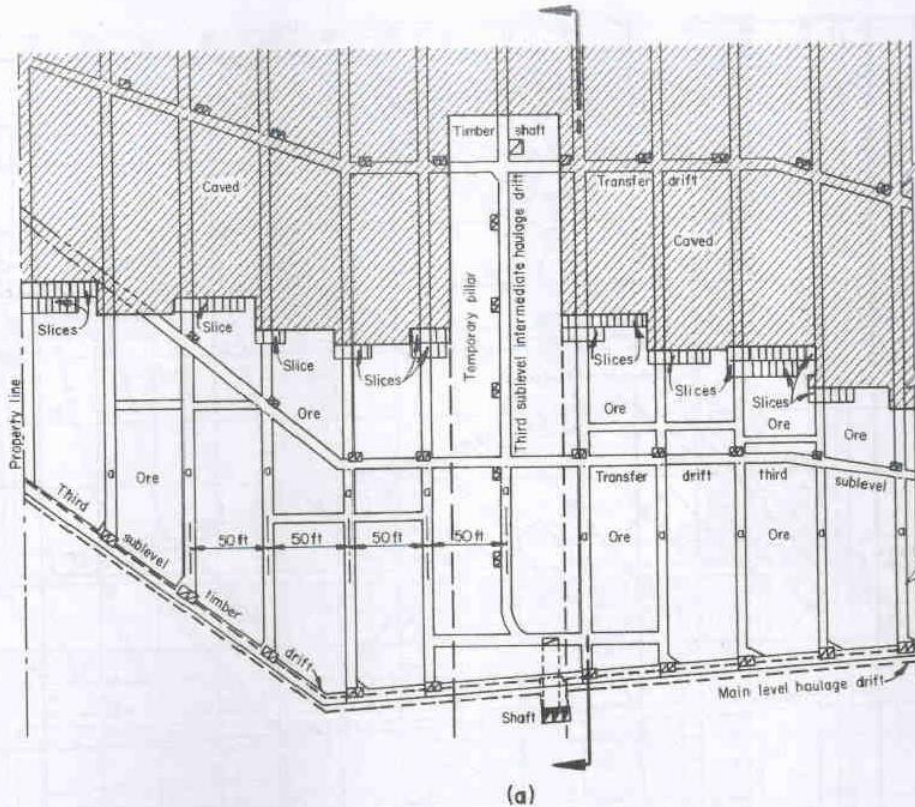


Fig. 8. Parallel top slicing, scraping to cars in a flat-dipping ore body two to four sublevels thick. (a) plan of part of the third sublevel (note a indicates tramming drifts); (b) cross section on X-Y.

Horizontal Parallel Top Slice114

Staggered face development at different level.

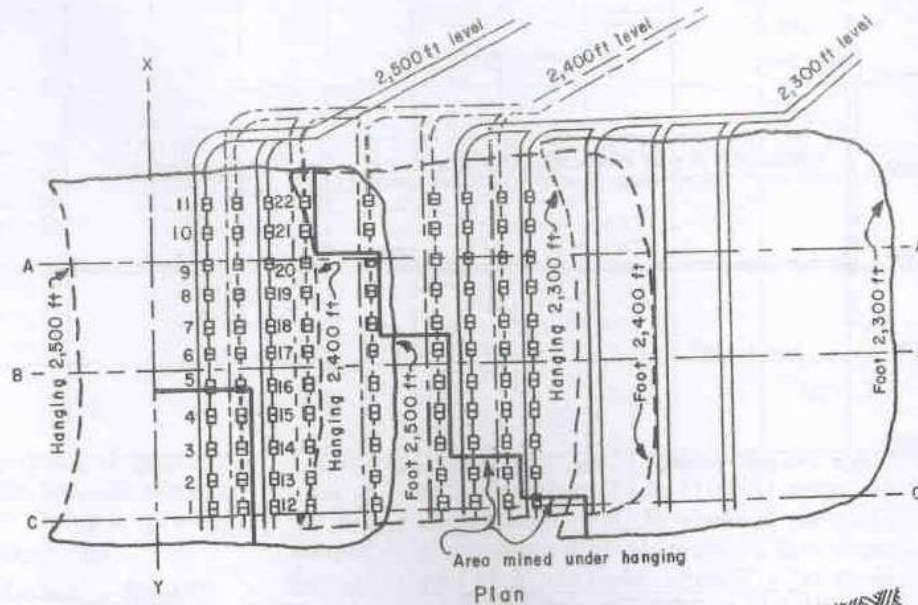
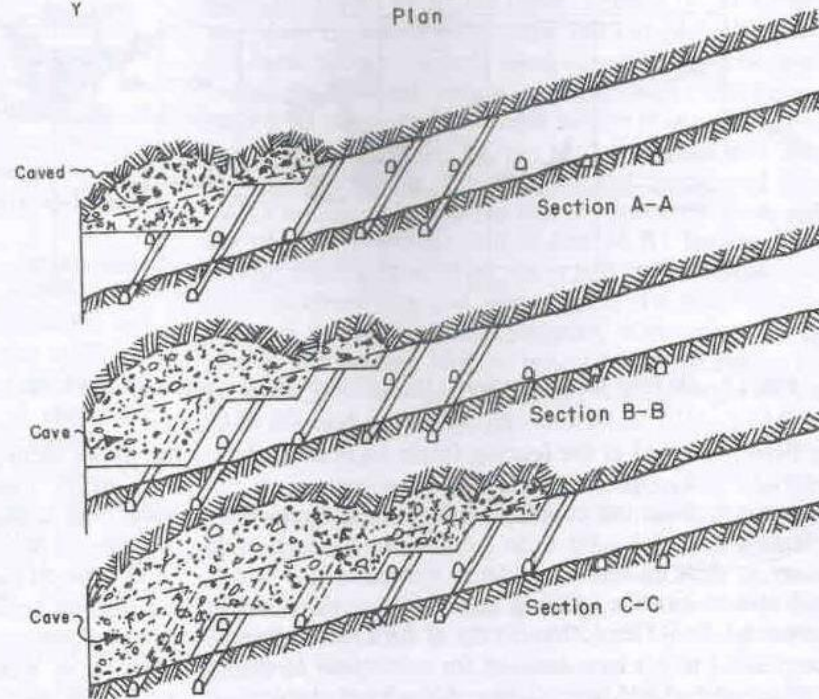
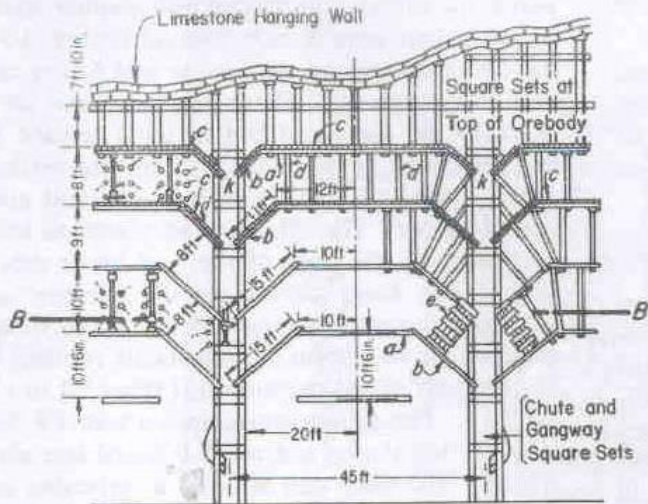


Fig. 20. System at the Athens mine (after Elliott). Metric equivalent: $\text{ft} \times 0.3048 = \text{m}$.

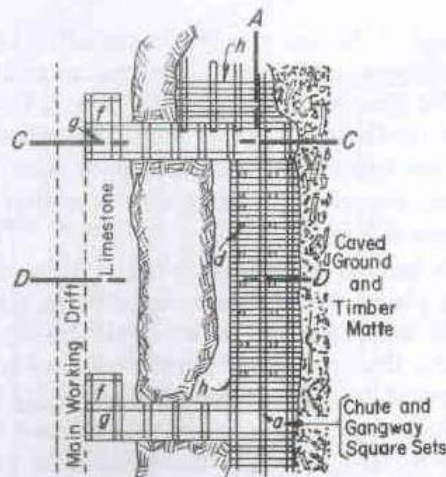


Horizontal Parallel Top Slice 115

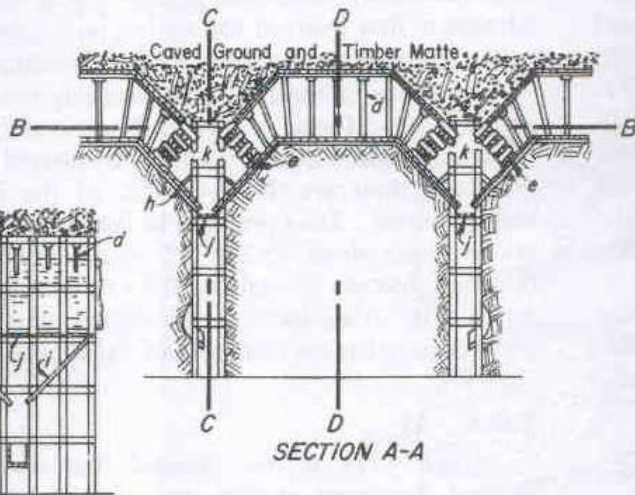
Slicing starts
below Squire
Set Stope.



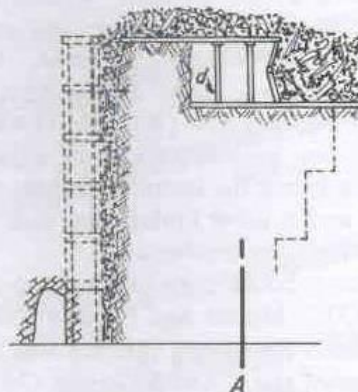
SECTION A-A
Showing Method of Timbering
and Pointing Holes



HORIZONTAL SECTION B-B



SECTION C-C



SECTION D-D

Fig. 21. Mitchell top-slicing caving system. (a) 101.6 × 254-mm (4 × 10-in.) sill; (b) 101.6 × 254-mm (4 × 10-in.) sill; (c) 50.8-mm (2-in.) floor on sills; (d) stull resting on ore; (e) bulkhead; (f) manway; (g) chute; (h) stope slide; (i) chute slide; (j) grizzly; (k) posts cut here. Metric equivalent: ft × 0.3048 = m.

Horizontal Parallel Top Slice116

Haulage
develop
ment.

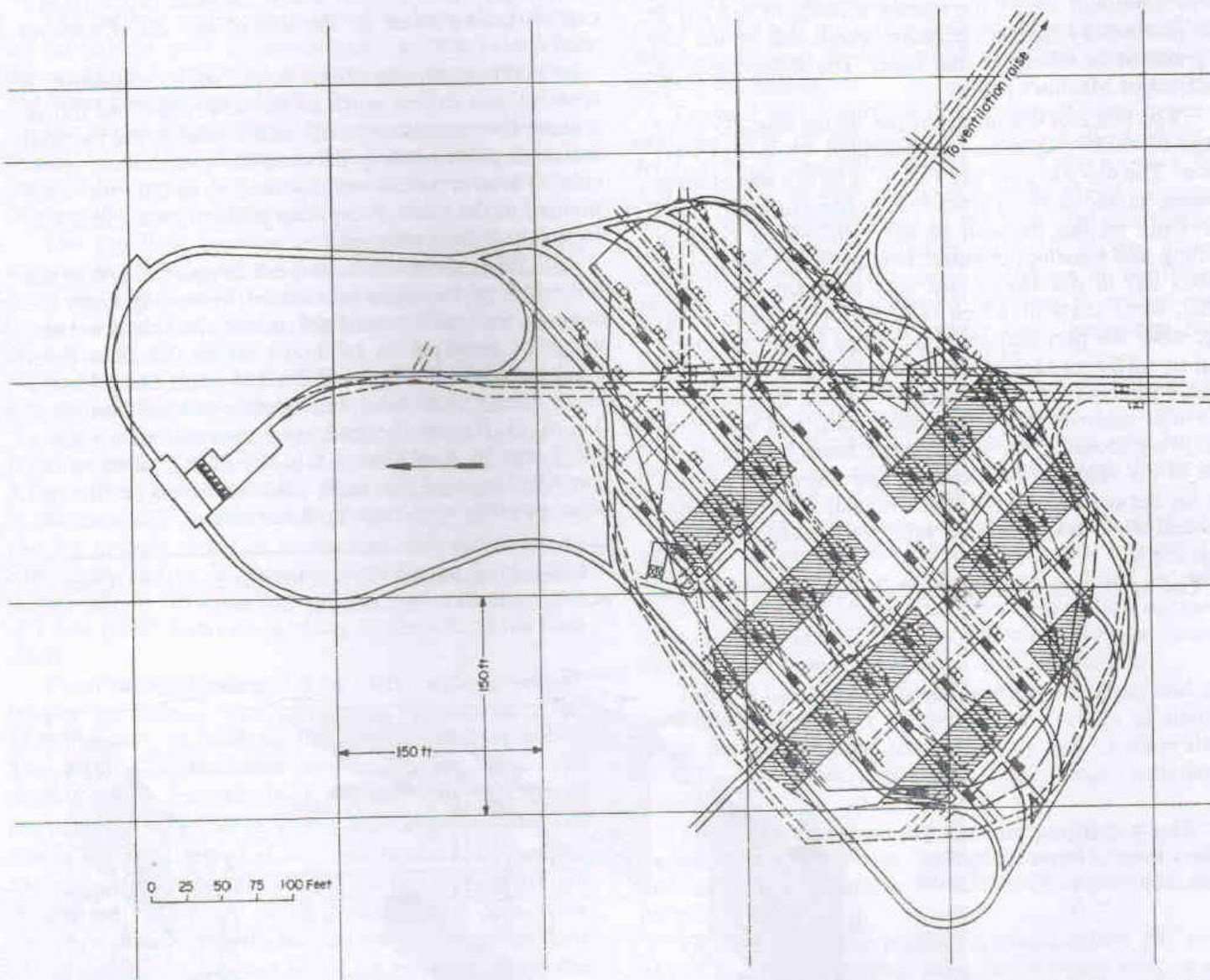
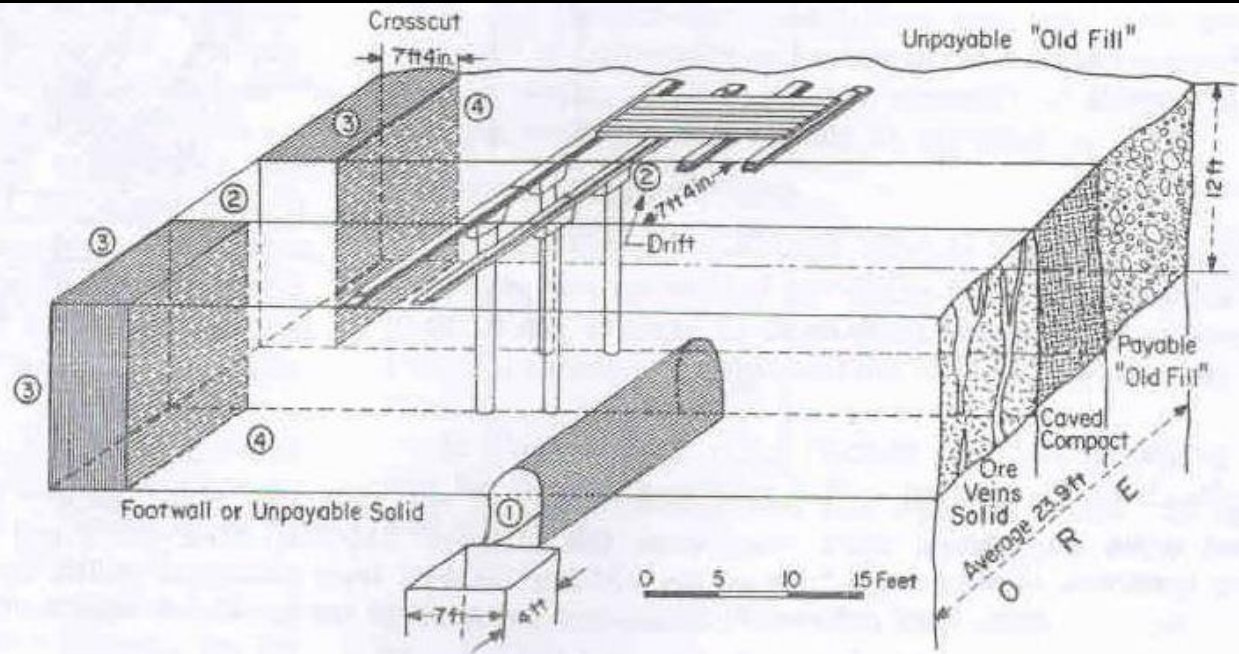


Fig. 22. Typical haulage and development levels at the Colorado mine. Broken lines show the 274.3-m (900-ft) level drifts and raises; solid lines show the 304.8-m (1000-ft) level drifts and raises; arrows from raises indicate branches; interior ore outline on the 274.3-m (900-ft) level shows the central core of waste; and the cross-hatched areas show cut-and-fill stopes laid out on both levels. Metric equivalent: $\text{ft} \times 0.3048 = \text{m}$.

Horizontal Parallel Top Slicing 117

Stoping in slid ore, payable old fill.

Fig. 23. Top-slicing at the El Bordo mine (after Mechin).
Metric equivalent: $\text{ft} \times 0.3048 = \text{m}$.



Radial Horizontal Top Slice 1

22A*In radial slicing ore is scrapped directly from face in to chute.

34BThick ore body is suitable for radial top slicing.

34B*In case of thick ore body radial top slicing with direct scrapping to chute is practiced.

34BFig 7,8,9

Radial Horizontal Top Slice 2

48A*In radial slicing, ore is scrapped directly in to chute with out any intermediate handling (Fig-5, 9)

48A*Radial slicing is designed to scrap ore from a considerable area directly in to chute with out dragging ore around sharp corner around.

Radial Horizontal Top Slice 3

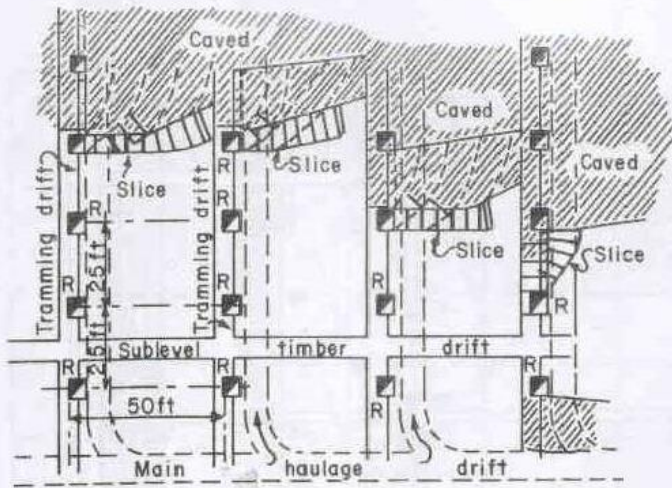
46A *Radial slicing is closely associated with scrapper loading.

46A *Radial slicing is best suitable for ore which is several sublevel thick.

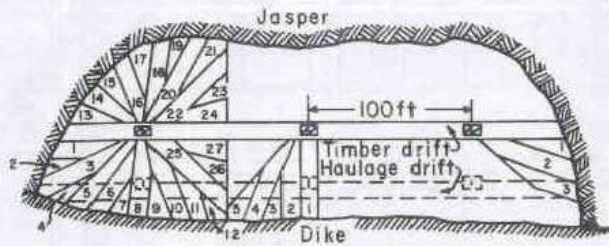
46A47*Radial slicing is best for ore body thick enough is that it is possible to arrange haulage drift with chute raise spaced optimally for economical scrapping distance & with out excessive development in waste rock.

48B*Post on side of next radial slice is removed.

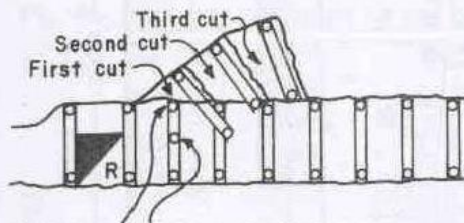
51A*Placing timber in case of radial slices is more difficult & complicated, than parallel slicing.



(a)



(b)



Put in center prop and remove
post before blasting first cut

(c)

Radial Top Slicing In To Chute 16

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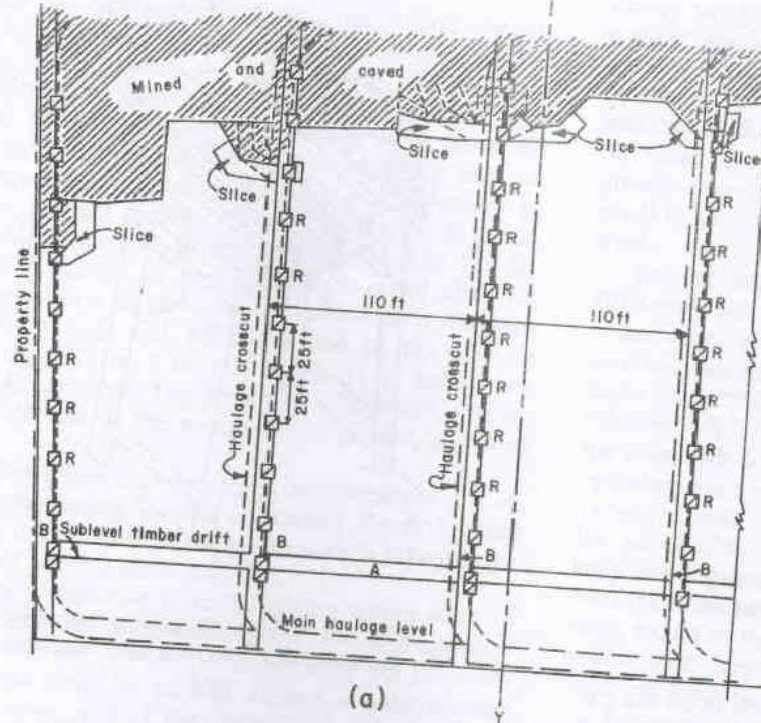
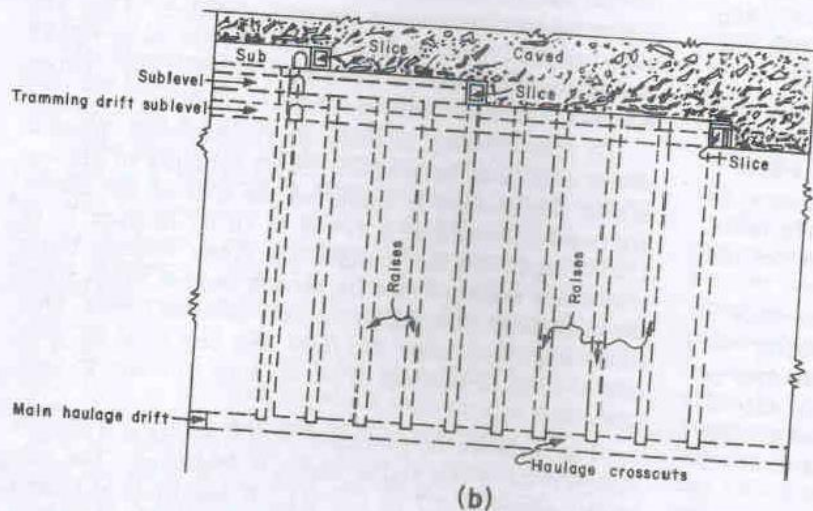


Fig. 9. Radial top slicing to chutes in a thick ore body. (a) partial plan of sublevel; (b) cross section on X-Y. Metric equivalent: $\text{ft} \times 0.3048 = \text{m}$.



Redial Top Slicing 112

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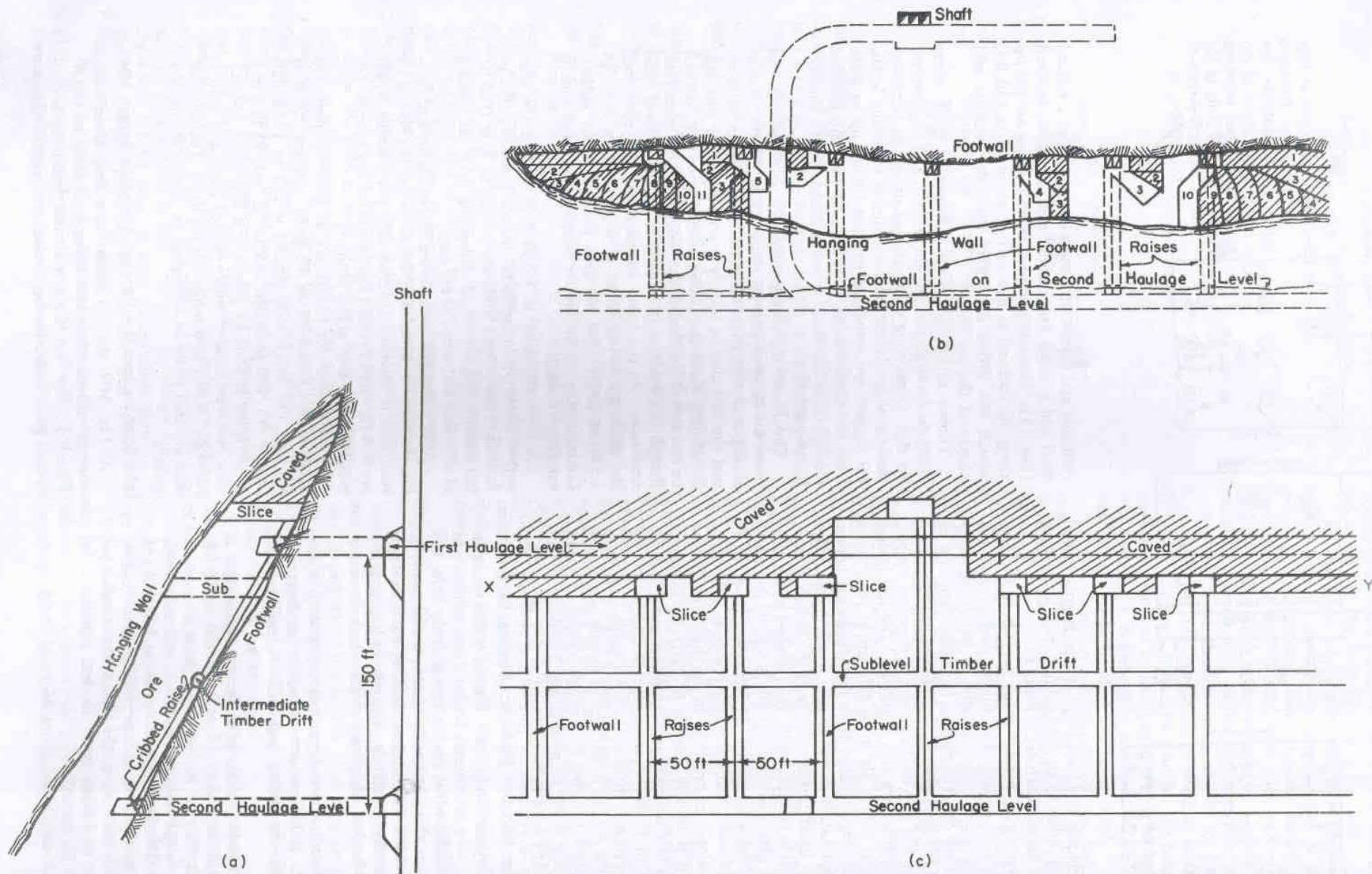


Fig. 15. Scheme of development and mining at Cuyuna Range. (a) cross section on line of shaft; (b) plan on X-Y; (c) vertical projection on footwall. Metric equivalent: $\text{ft} \times 0.3048 = \text{m}$.

Incline Top Slice 1

23A*In incline top slicing, ore is taken to chute with help of gravity.

86A*Incline top slicing eliminates ore shoveling on stope.

86A*Broken ore is blasted down on slopping floor of stope.

86A*Broken ore runs in to narrow shrinkage stope & then bottom of trough.

Incline Top Slicing 2

86A*Ore from shrinkage stope was drawn directly into car through chutes on motor haulage level.

86A*OMS of incline top slicing is 12 in compared with flat top slicing 5.

87A*Haulage sublevels are at 16m vertical apart.

87A*Raise are excavated from sublevel to stope.

87A*Each sublevel has central drift through ore body.

Incline Top Slicing 3

87A*Drift has cross cut at right angle in 16m interval.

87A*From drift un-timbered raise of 1.2m dia is placed in every 5m interval.

87A*Line of raise from a center line of individual slices.

87A*Stoping incline (drift) is driven from 14m above sublevel.

Incline Top Slicing 4

87B*This heading starts of from top of raise.

87B*Inclines headings are timbered with 3m & 0.7m apart.

87B*Stoping operation starts after incline heading to intersect four raises.

87B*Stopes are divided in panel of 3m width 7 16m long (same distance between two crosscut at sublevel).

87B*Floor of panel is inclined at 33 degree up on either side of center heading.

Incline Top Slicing 5

87C*Floor of heading is funneled down to raise.

87C*Blasted ore from face runs down in to raise without any handling.

87C*When heading had advanced up to open end of raise, timber of panel (slice) is blasted & another slice starts.

87D*Post in panel are of 200mm (8inch) round timber, 3m long.

87D*These posts were not set normally to floor of panel, but inclined to wards center heading at an angle of 16 degree from vertical.

Incline Top Slicing 6

- 87D*Flooring consists of 50mm X 300mm plank laid up on sill of round timber.
- 87D*Sill are so placed as to be readily picked up by post of next lower slice.
- 87E*Ore is drawn from chute on sublevel & hand trammed to motor haulage pockets.
- 87E*Timber consumption is less in incline top slicing.

Incline Top Slicing 7

87E*Total reduction of cost is about 15% in compared to normal top slicing.

87E89*Handling & installing timber, drilling equipment & other machinery is difficult in stope is a disadvantages.

89E*Top slicing method of working is mostly replaced by Sub level caving or Cut & Fill method.

Incline Top Slicing 11

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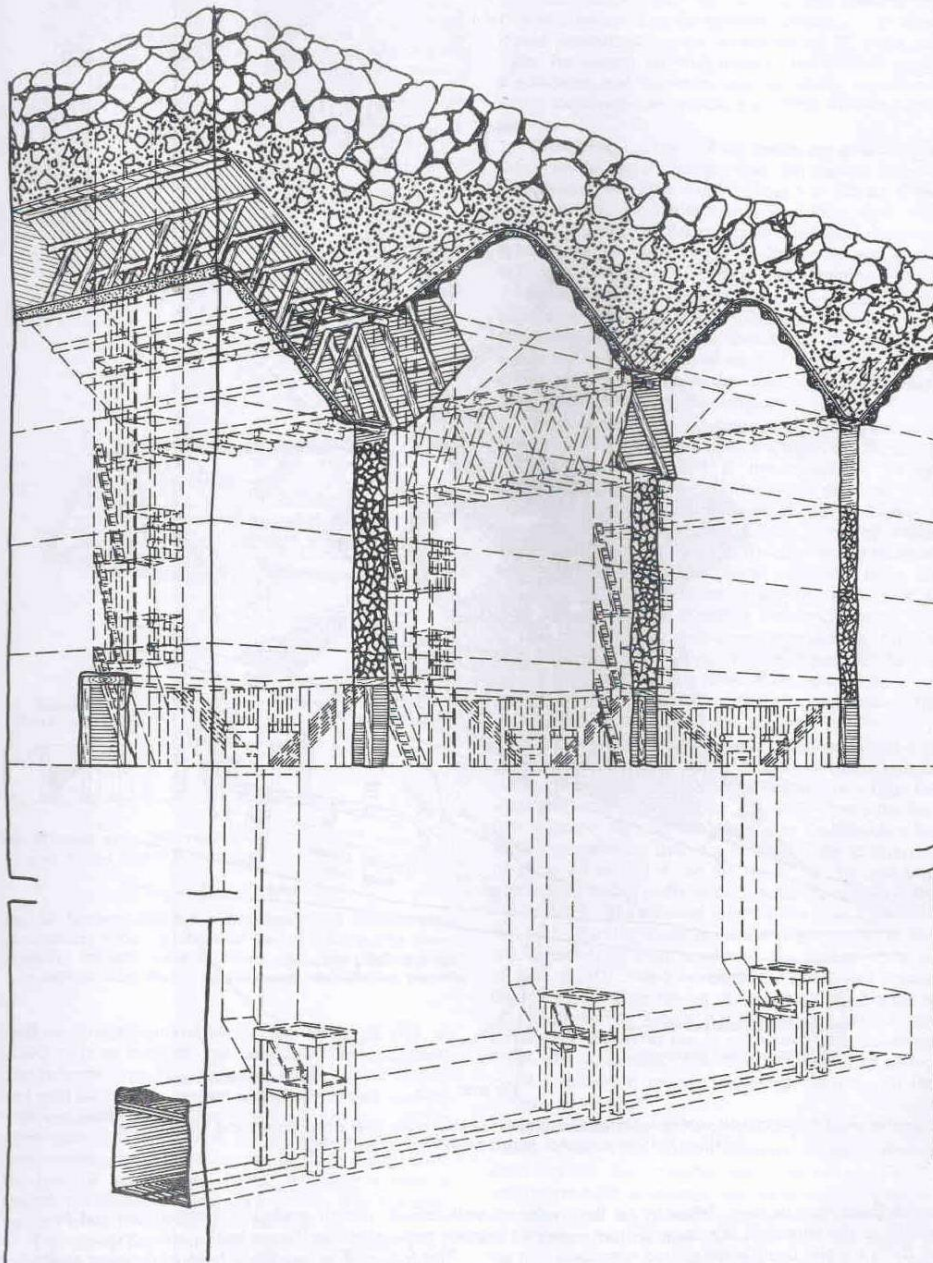


Fig. 3. Perspective sketch of Coronado incline top-slicing mining method.

Production Capacity

15A*For higher production rate, more number of working places are to be opened.

15A*Working places are scattered over a considerable area.

15A *These working places are connected by large & long development opening.

*

Caving 1

19A*Some time capping braeks in large blocks which wedge together so as to leave a large open space below them.

19A*Sudden collapse presents a serious element of danger to slicing operation below.

19A*Large open space above Gob on top of active slice is a very bad condition for stope.

*After a reasonable progress in extraction of the top slice the roof is induced to cave in by removal of timber support and rest on the spread-out timber mat.

Caving 2

27A*Some time, when capping is hard, but jointed, which allow it to break & cave.

27A*In this condition, two or three slices are to be taken before caving starts.

27A*In case of presence of quick sand or other similar material, it is better to leave one meter ore against sand for first slice.

27A*Hay or straw can be packed over ore logging, before blasting to prevent quick sand to run down on slice.

2A*A strong hang wall shall prevent proper caving due to arching over of blocks of broken caving between

Caving 3

2A* A strong hang wall shall prevent proper caving due to arching over of blocks of broken caving between the walls.

1A* A capping which breaks in to large block which arch over a hang up so as to have large open space in cave constituent source of danger in top slicing.

70A* Some time slab in gob are to be blasted for full complete caving in stoping system.

70A* Holes are drilled in to slab, in case of soft slab, hand auger drill is proffered because it causes less vibration.

Timber Mat

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Drilling Blasting 1

70A*Drilling in ore body (ore face) is done with light jack hammer drill.

70A*In side all holes, expect bottom lifter, are drilled at upward angle (Fig – 14).

70A*For hard rock development, relatively hard or medium weight drifters is deployed..

Drilling Blasting 2

71A*In slice, powder factor is 0.2kg (explosive)/ton of ore.

71A72*In top slice care must be taken to avoid blasting with through to wards timber.

71A*Normally last set of timber is very close to face.

71A*In hard, tough ground, damaging due to blasting is common.

71A*Before blasting collar brasses are put in between caps of last two sets.

71A*Some time cap is reinforced by tie brasses to legs.

71A *This usually prevent knocking down timbers.

Drilling Blasting 2

72B*Some time timber is broken, if side & back holes are too hard loaded with explosive.

72B*Some time first (top) hole do not break well. Then lower hole throws back rock to ward support.

79A*Some time capping hangs up over considerable area.

79A*In this case, holes are drilled in cap & blasted down.

Blasting 110-A

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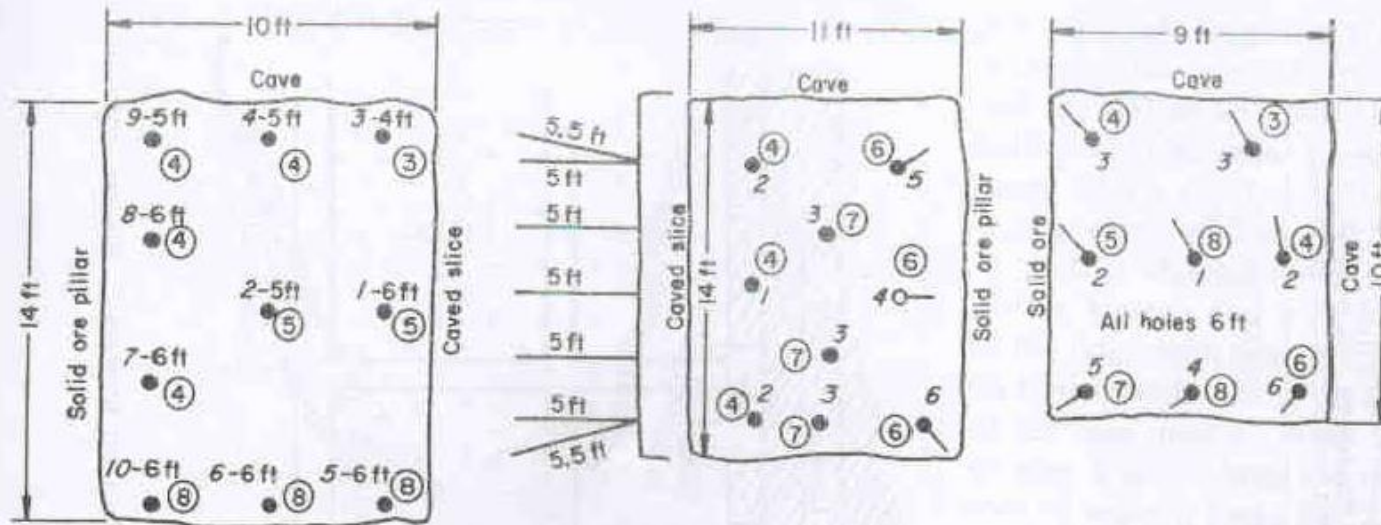


Fig. 14. Types of top-slice rounds. Numbers in circles indicate the number of sticks of dynamite; other numbers indicate the order of firing. Metric equivalent: $\text{ft} \times 0.3048 = \text{m}$.

Support 1

- 39A*Top slice usually advances 1.5m each round & slice timber are set on 1.5m center as slice advances.
- 39A*Slice timber consist of simple drift set of round timber, with “Joggle” cut on end of cape. Fig 10.
- 39A*These sets are placed & blocked to side of slice (timber on cave side & ore on opposite side).
- 39A*Lagging usually of round pole or split round timber, turned split logging.

Support 2

40A*After blasting & before mucking out face of slice, two, three or more round of poles are driven over cap of last standing set & under cap of next set, fore-poll fashion (Fig-11).

40A*These poles cross lagged with short boards or pieces of lagging as a protection to men working at face & to catch up overlaying mat or capping.

Blasting Down Support 1

38A*Each support of each slice is blasted down when slice is completed.

63A*When a slice is completed, side next to solid ore covered is usually boarded up or covered with wire fencing & bottom is covered down.

63A*When sand or some other loose material is present in gob, it is necessary to cover bottom & side with board to prevent contamination of ore in next slice or in next sublevel below.

63A*When there is no ore below, as in single sublevel mines or when bottom slice of ore body has been reached, bottom is not required to be covered.

Blasting Down Support 2

- 63B*Some time wire fencing & board both are used for side & bottom.
- 63B*When wire fencing is used for bottom, first poles are laid longitudinally, then covered by wire fencing.
- 63B*Wire fencing is secured with pole by few staplers.
- 63B*Poles are spaced at 0.3m to 0.6m apart & 5m in length, enough to span two sets.
- 63B66*When there is sand in gob, poles are laid fast & then covered with board & wire fencing.
- 63B*Mining board are hard wood 10mm to 13mm thick.

Blasting Down Support 3

66B*Wire fencing is made of low priced galvanized wire, diamond mesh, station guard fencing, 1m in width & 50m long in roll..

66B*Mesh is strengthened by 11 longitudinal strand of double wire twisted together.

66B*Cost of handling & placing wire is much less in compared to board.

66B67*For deep mine, thick timber mat is used.

66B*In deep mine, it is not necessary to cover side skice as blasted down gob is practically wood.

Blasting Down Support 4

68B*After completing slice, post next to cave are drilled with augur & blasted down with explosive.

68B*Next slice is started either next shift or next day.

68B*In case cave is not tight & still active & working, a short lapse of time is desirable before starting next slice.

68B69*Similarly all slices are excavated, slice starts in next sub level below cave.

68B*Slicing thus proceed on several sublevel at same time.

Blasting Down Support 5

69B*Slicing on any given sublevel must always lag behind that on sublevel above by a few slice, so that mining is always under a settled mat.

Timber 1

- 3A*Top slicing requires use of considerable timber & logging.
- 3A*It is essential that plantifull of cheap timber be available.
- 3A*Timber supply must be assured & price should be low in compression to value of ore to be mined.
- 3A*Slice timber is required to stand only for a relatively short time & then blasted out.
- 3A*A some what poor grade of timber is acceptiable than would be for permanent support.

Timber 2

- 3B*Timber should be tough, capable of under going considerable crushing & bending before breaking.
- 3B*Round green timber with bark on is usually deployed.
- 3B*If an adequate supply of timber is not available or its cost is too high in relation to value of ore, some other suitable method of working gas to be deployed.
- 18A*Handling of timber, lagging, boards & wire fencing is expansive in top slicing.
- 18A*Theses, material in large quantity has to be handled long distance in a mine.

Timber 3

42A *Round green timber with bark left on is used for “Slice Set”.

42A 43*Timber purchased on specification as to size of top or smaller end of timber.

42A *Timber requirement is different for different mine, depending up on width, height of slice, pressure developed, size & price of timber available.

42A 44*In shallow mine, where load (pressure) is not heavy, post are set with little if any batter.

Timber 4

44 *In deep mine, where there is heavy pressure & especially where much sited pressure develops, post will battered. Fig 10

44 *Caps are formed (Fig-10) to have a full bearing on post.

44 *Some time, end of cap extended even beyond top of post.

Timber Support In Slice18

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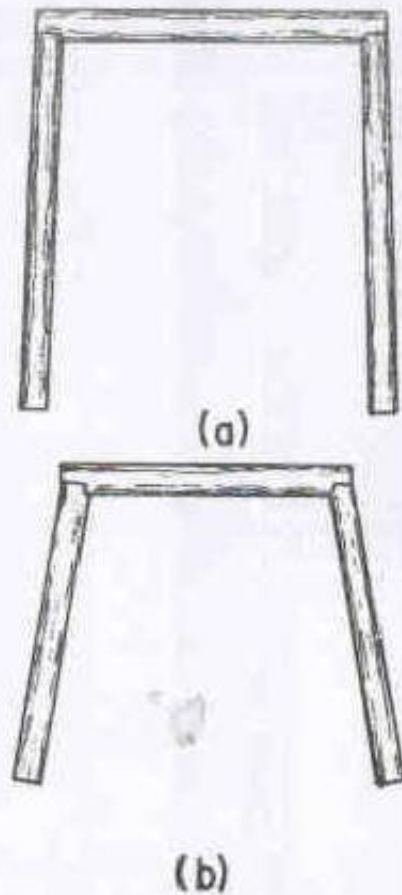


Fig. 10. Slice sets. (a) set for shallow depth and light gob; (b) set for heavy gob.

where there is heavy side pressure, posts are battered 5 mm (2 1/4 in.) to the foot. At some mines

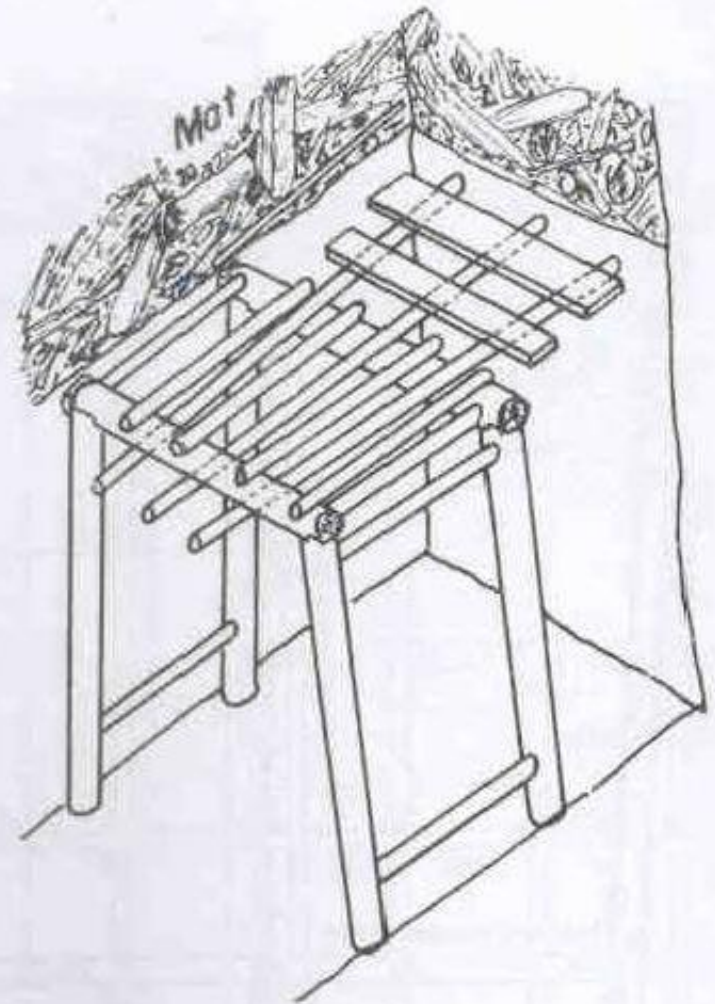


Fig. 11. Slice timbering showing forepoles for catching up overlying mat.

Loading Hand 1

37A*In case of hand loading of ore in to car, two slices are taken side by side, with one set of track laid.

37A*Width of slice becomes 2.4m (1.2m+1.2m.)

37A*Some time three slices are taken, when support is able to take weight.

Loading Scarper 2

38A*In case of loading with scarper, standard width is 3m.

38A*In case of two slice is taken together, it is difficult for scrapper to reach entire width of slice.

38A*Wider slice support also need a center prop to set under cap, causing obstruction to scarpper.

Loading Scraper 3

- 54A*Ore from slice is shoveled by hand in to sublevel car & tramping to main level chute.
- 54A55*Ore from development & slicing is handled by power scrapper, which either load in to sublevel car or scrap directly in to chute.
- 54A*A timber slide is constructed to guide ore ion to car.
- 54A*Loaded scrapper is pulled up slide & ore is discharged in to car placed underneath.
- 54A*Scrapper hoist is usually placed in line with drift on opposite side from slice (Fig-12).

Loading Scraper 4

55A*Some time scrapper hoist is put on staging back in drift, in which case sheaves must be employed to guide rope around corner.

55A*When scarping direct to chute, no special preparation of place is required.

55A*In direct scrapping, radial or angle slicing often employed instead of parallel or right angle slicing to avoid pulling scrapper round sharp corner.

55A*For pulling empty hoe back to face, a trial sheave is required at face.

55A56*Return pulley is fixed at face with a boom, which permits cleaning face well after each blast.

Loading Scraper 5

61A*Scraper with single drum.

61A*In single drum scraper, scraper hoe has to be manually pulled back to face manually.

61A*Single drum scraper can be used up to 9m distance.

61A*Double drum scraper with hoe or box.

61A*In case of double drum scraper, empty hoe goes back to muck pile by scraper it self.

61A*It is difficult to move bigger equipment through raise, timber drift, hence high level of mechanizations not possible.

Loading Car With Slide 19

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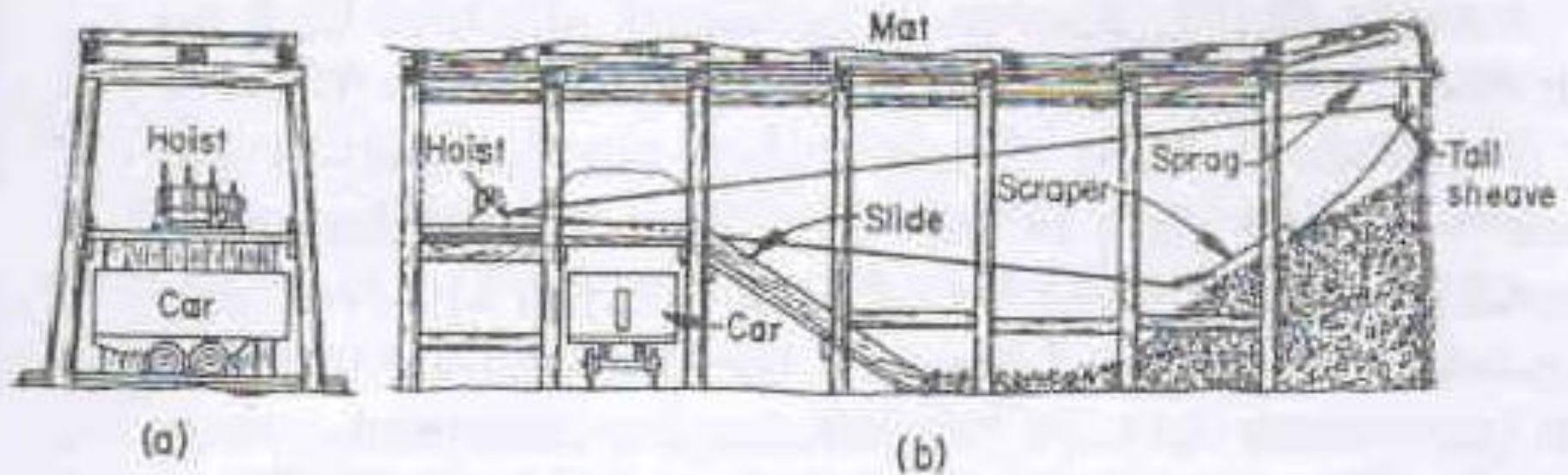


Fig. 12. Scraper loading from slide to cars. (a) rear elevation, (b) side elevation.

Return Sheave 111-A

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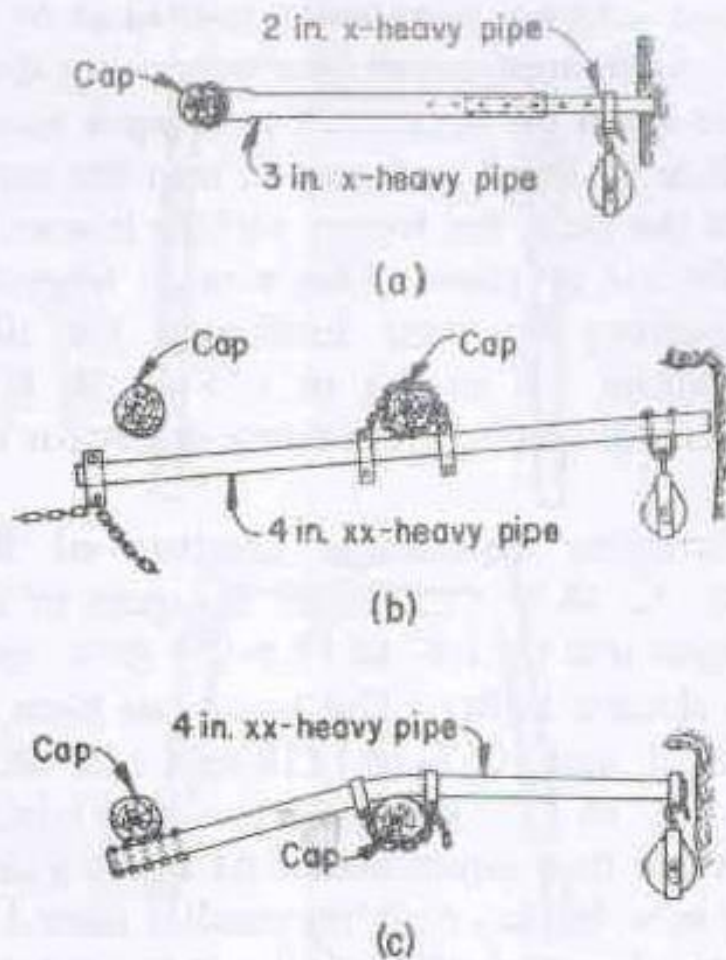


Fig. 13. Types of booms for sheave support. (a) telescoping boom; (b) swing boom, rear and end anchored by chains; (c) swing boom, rear and end anchored by teeth. Metric equivalent: in. $\times 25.4 =$ mm.

Ventilation

- 13A*Ventilation is difficult in top slicing & in deep mines working under thick mat of timber.
- 13A*In heavy sulfide ore which is oxidized, when exposed to mine air with generation of heat & possibility of gas, ventilation is an important consideration.
- 13A*In some cases, small booster fan is established.
- 13A*Forced ventilation from surface fan of large capacity is used in deep mine.

Lighting

63A*Slice is generally illuminated by electric light.

63A*Wire & other electrical fittings are heavily insulated for under ground use.

Fire Hazard

- 14A*Due to accumulation of large amount of old timber in top slice operation, a possibility of fire hazard is created.
- 14A*Top slice method of working is deployed in relatively weak & jointed formation.
- 14A*This results in large water percolation, keeping old timber wet.
- 14A*Though, there is lot of accumulation of timber, cases of fire is not common.

Advantages 1

7A High extraction of ore with very low dilution from capping & walls.

*Top slicing is a high recovery method.

8A* Where market condition is such that occupational shut down of stope or slices may be required, stopes or slices may be blasted so that ore body remain in good condition for resuming production operation on short notice, even after idleness of considerable duration.

*Top slicing method can be deployed, where over head method can not be practiced.

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Advantages 2

- *Top slicing can be employed for ore body under sand or other loose material
- 6A*Top slicing is a safe & economical method of mining, ore body, which can not be economically, safely mined by over hand stoping method.
- 10A*Top slicing can be employed under sand & other loose surface material, with least dilution from over laying sand or loose material.
- 12A*Under deep mining condition where ore body is thin, it can be employed with out causing subsidence at surface.

Disadvantages 1

- *Production rate is low.
- *Cost of production is high because of high labour & timber consumption.
- *Top Slicing can not be mechanized.
- *Top Slicing method can not achieve high rate of production.
- *Top Slicing Sublevel caving gives high exposure of man under caved roof.
- *Top Slicing Sublevel caving is difficult to ventilate.

Disadvantages 2

- *Dry timber along with sulfide can be source of fire.
- *If roof does not collapse for considerable time, sudden collapse of roof shall cause air blast as well as damage to slice below.
- 11A*If timber & logging are not in adequate supply & are expansive, top slicing method can not be deployed, even other method has lower ore extraction & higher dilution of ore.
- 12A*Top slicing can not be deployed where surface subsidence is a problem.

Disadvantages 3,

*12A*Some time fault plane , jointing or other weakness in capping may causes subsidence at some distance from area immediately over mine.

8A*Development heading may require considerable repair work & maintenance, if close to caved ground.

References I

*SME, Underground Mining Methods Hand book, W
A Hustrulid, Page 758 to 789